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Bi-level coordinated expansion planning of high-PV-penetration active distribution networks incorporating multiple flexible resources considering WGAN-GP-based uncertainty generation

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ABSTRACT

Integrating high-penetration random photovoltaics (PVs) into the active distribution network (ADN) often causes overvoltage issues, which can even trigger ADN failures. To accommodate high-penetration PVs integration while maintaining economic efficiency and flexibility of ADN, a bi-level collaborative expansion planning model considering uncertainty generation and reduction is proposed in this paper. Based on described uncertainty scenarios of PV and load obtained by proposed Wasserstein Generative Adversarial Network with Gradient Penalty (WGAN-GP) and K-means method, the upper level firstly configures soft open points to adjust the ADN topology, then optimizes the siting and sizing of PVs, electric vehicle charging stations, energy storage systems, and active management resources (on-load tap changers, and static var compensators) with minimum investment cost. Lower level uses multi-objective mixed-integer second-order cone programming (MISOCP) model with network loss, PV curtailment and voltage deviation of different attributes to optimize ADN scheduling. To solve this planning model, a hybrid algorithm that integrates an adaptive particle swarm optimization algorithm with SOCP is proposed, employing the technique for order preference by similarity to an ideal solution (TOPSIS) to address the multi-objective issue. Finally, the presented model was tested in the actual rural 41-bus and urban 40-bus ADN of Gansu, China, demonstrating that total costs decreased by 14.4 % and 12.5 % respectively compared to plans without active management resources.

Nomenclature

Abbreviation	
ADN	Active distribution network
DNO	Distribution network operator
ESS	Energy storage systems
EVCS	Electric vehicle charging station
MISOCP	mixed-integer second-order cone programming
OLTC	On-load tap changer
PV	Photovoltaic
RES	Renewable energy source
SOCR	Second-order cone relaxation
SOCP	Second-order cone programming
SOP	Soft open point
SVC	Static var compensator
Sets and indices	
$(i, j)/\Omega$	Index/set of MV buses

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$h/\ \Omega_h^i$	Index/set of LV buses and relevant MV/LV transformer units connected to MV bus i
k	Index of sample variations of EV charging stations
$l/\ \Omega_l^{ij}$	Index/set of lines where i is the sending bus and j is the receiving bus
s/S	Index/set of uncertain scenarios
t/T	Index/set of hours
Parameters	
A_i^{SOP}	Loss coefficient of SOP at node i
B_l	Shunt susceptance of line l
C_{ADN}	ADN investment construction cost
$C_{PV}, C_{EVCS}, C_{ESS}$ $C_{SOP}, C_{OLTC}, C_{SVC}$	Investment construction cost of PV, ESS, EVCS, SOP, OLTC and SVC
$c_{cons}^{PV}, c_{cons}^{EVCS}, c_{cons}^{ESS}$ $c_{cons}^{SOP}, c_{cons}^{OLTC}, c_{cons}^{SVC}$	Installation costs of unit capacity of PV, ESS, EVCS, SOP, OLTC and SVC
C_{opt}^{a}	Annual operating cost of distribution network

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(*continued*)

C_b^a, C_p^a, C_m^a	Annual cost of buying and selling electricity, PV curtailment and maintenance cost of ADN operation
c_b, c_e	Unit price of purchasing and selling electricity to the superior distribution network
c_p	Unit cost of PV curtailment
$\overline{E}$	Unit energy capacity of ESS
E_{ini}^{ESS}	Initial energy state of ESS unit
g_j, b_j	Conductance and susceptance of bus j
l_{loss}^a	Annual network loss of ADN
$\widetilde{I}_{\min}, \widetilde{I}_{\max}$	Minimum and maximum limited current square
$m_{pv}^a, m_{EVCS}^a, m_{ESS}^a$ $m_{SOP}^a, m_{OLTC}^a, m_{SVC}^a$	Unit annual maintenance cost of PV, ESS, EVCS, SOP, OLTC and SVC
P_{PV_rated}	Rated power of unit PV generation
P_{EVCS_rated}	Rated power of unit EVCS
Q_{SVC_rated}	Rated power of unit SVC
S_{SOP_rated}	Rated power of unit SOP
$\overline{P}_{ESS}$	Unit charging or discharging power upper limit of ESS
$P_i^{p\_load,MV}$	Peak of load demand directly connected to MV bus i
$P_{h,i}^{p\_load,LV}$	Peak of load demand at the LV bus h of MV bus i
r_l	Resistance of line l
$\partial_i^{min}, \partial_i^{max}$	Minimum and maximum the square of OLTC adjustable ratio
S_l	Flow limit of branch l
$TR_{h,i}^{lim}$	Rated power of MV/LV transformer between MV bus i and LV bus h
V_d	Total voltage deviation of distribution network
V_{mean}	The average node voltage
$\widetilde{V}_{\min}, \widetilde{V}_{\max}$	Minimum and maximum limited voltage square for MV buses
x_l	Reactance of line l
y	Service life of module installed to network
γ_s	Total days of scenario s throughout the year
$a_{s,t}^{PV}$	Scenarios of PV generation during uncertainty description
$a_{s,k,t}^{EVCS}$	The k-type sample of EV charging trend in period t of scenario s
ρ	Maximum PV integration ratio permitted for transformer
ζ	Minimum photovoltaic penetration for distribution network issued by management
λ^{ch}, λ^{dis}	ESS charging and discharging efficiency
ε	Discharge depth of ESS units
Decision variables	
Ω^{PV}, Ω^{ESS}, Ω^{EVCS}, Ω^{OLTC}, Ω^{SVC}, Ω^{SOP}	Candidate Set of PV, ESS, EVCS, OLTC, SVC installation notes and SOP construction
$\Omega_h^{i,PV}$	Set of candidate PV installation notes of LV bus h connected to MV bus i
$I_{s,l,t}, \widetilde{I}_{s,l,t}$	Current and square current of line l in period t of scenario s
$n_{MV,PV,i}, n_{LV,PV,i,h}$	Number of reference PV panels installed within MV bus i or LV bus h of MV bus i
$n_{EVCS,k,h,i}, n_{ESS,h,i}$	Number of EV charging station and ESS installed within LV bus h of MV bus i for sample EV load k
$n_{SVC,i}, n_{SOP,i}$	Number of reference SVC and SOP installed within MV bus i
$E_{s,h,i,t}^{ESS}, E_{s,h,i,t-1}^{ESS}$	State of energy of ESS units connected to LV bus h of MV bus i in period t or period *t-1* of scenario s
$P^{cap,PV}$	Total PV capacity connected to distribution network
$P_i^{cap,PV,MV}, P_{h,i}^{cap,PV,LV}$	PV capacity directly connected to MV bus i and LV bus h of MV bus i
$P_i^{cap,EVCS}, P_i^{cap,ESS}$ $Q_i^{cap,SVC}, P_i^{cap,OLTC}, S_i^{SOP}$	Total EVCS, ESS, SVC, OLTC and SOP capacity integrated to the bus i
$P_{s,l,t}, Q_{s,l,t}$	Active, reactive power flow at receiving end of line l in period t of scenario s
$P_{s,t}^{Sub}, Q_{s,t}^{Sub}$	Active transactive power through substation node in scenario s and hour t
$P_{s,i,t}^{SOP}, Q_{s,i,t}^{SOP}$	Active/reactive power generated by SOP node i in scenario s and hour t
$P_{s,l,t}^{SOP,loss}$	SOP transmission loss of line l in scenario s and hour t
$P_{s,i,t}^{PV,MV}$	PV power available at the MV bus i in period t of scenario s
$P_{s,j,t}^{G}, Q_{s,j,t}^{G}$	Injected active, reactive power at bus j in period t of scenario s and hour t

(*continued on next column*)

(*continued*)

$P_{s,j,t}^{L}, Q_{s,j,t}^{L}$	Received active, reactive power at bus j in period t of scenario s and hour t
$P_{s,h,i,t}^{PV,LV}$	PV power available at the LV bus h in period t of scenario s
$P_{s,h,i,t}^{ESS,ch}, P_{s,h,i,t}^{ESS,dis}$	ESS charging and discharging power at the LV bus h of MV bus i in period t of scenario s
$P_{s,h,i,t}^{PV,LV}$	Total PV power available at the LV bus h of MV bus i in period t of scenario s
$P_{s,h,i,t}^{L,MV}$	Load demand at the bus i in period t of scenario s
$P_{s,h,i,t}^{L,LV_{other}}$	Other load at the LV bus h of MV bus i in period t of scenario s
$P_{s,h,i,t}^{EVCS,LV}$	Total EV charging power by LV bus h to the MV bus i in period t of scenario s
$P_{s,h,i,t}{}^{LV+}, P_{s,h,i,t}{}^{LV-}$	Total power injected/drawn by LV bus to MV bus i in period t of scenario s
S_p^{max}, S_q^{max}	Upper limits of active and reactive power exchanged with the upstream grid
$Q_{s,i,t}^{SVC}$	Reactive power compensation of SVC at the bus i in period t of scenario s
$\partial_{s,i,t}$	Discrete value variable for the ratio of the secondary side to the primary side of OLTC
$u_{sub,s,t}$	Binary variable for transactive power through substation node of hour t in scenario s. Power obtained from the superior network is 1, otherwise 0
$u_{s,h,i,t}^{ESS}$	Binary variable for ESS model at the bus i of hour t in scenario s. 1 if ESS model is charging, otherwise 0.
$u_{s,i,t}^{OLTC}$	Binary variable for OLTC action of the node i at period t in scenario s. 1 if the OLTC increases the voltage, otherwise 0.
$V_{s,i,t}, \widetilde{V}_{s,j,t}$	Voltage magnitude and voltage square of the node i at period t in scenario s
$u_{s,h,i,t}^{LV}$	Binary variable of logical constraints for the power decomposition of LV bus h

1. Introduction

Vigorously developing renewable energy source (RES) and increasing the installation capacity of RES are key to China's energy transition and hold significant strategic importance. A large-scale integration of distributed photovoltaics (PVs) into active distribution networks (ADNs) is an inevitable trend. In the process of advancing "County-Wide Photovoltaics" projects in China, to meet the demands of high photovoltaic penetration, it is often necessary to rent residential rooftops and collaborate with distribution network operators (DNOs) to construct PVs. The interest generated from selling electricity is then used to subsidize the residents. Moreover, new loads such as electric vehicle charging stations (EVCSs) has been integrated into ADNs. Their continuous integrations on both sides of source and load increase uncertainties throughout the ADN by changing the voltage distribution and power flow direction [1–3], which greatly reduces stability and creates significant challenges for ADN expansion planning (ADNEP) [4,5], even the ADN cannot accommodate high-penetration PV integration. DNOs urgently need a long-term planning strategy for improving network flexibility while maintaining economic efficiency [6].

DNOs have considered integrating energy storage systems (ESSs), on-load tap changers (OLTC), and static var compensators (SVCs) to enhance the operational flexibility of ADNs [7]. On the demand side, demand response (DR) is a mean of mitigating the uncertainty and intermittency of renewable generation [8]. Especially with the increasing demand for EVCSs, providing guidance for users with charging needs helps with the large-scale integration of PVs into an ADN. Planning investment in PVs and integrating new technologies such as ESSs and EVCS to comply with policies set by higher levels of government to encourage PV penetration is a widespread problem [9].

In recent years, with the assistance of these active management resources, a great variety of studies have focused on extensive integrations of RES in ADN. Reference [10] considers a practical situation that the siting choice of individual RES owners could be conflict with systems operation, a trilevel ESS planning formulation with "min-max" risk constraint is developed. Reference [11,12] research the co-planning of

Table 1
Summary of previous studies in distribution network planning.

Reference	Multiple flexible resources with PV					Multi-objective		Uncertainty	Optimization methodology
	ESS	EVCS	SVC	OLTC	SOP	Ec.	Se.		
[7]	✓		✓	✓	✓	✓			MISOCP by CPLEX solver
[9]	✓	✓				✓			SOCP by CPLEX solver
[20]	✓	✓			✓	✓			MISOCP by solver
[34]	✓					✓		PDF sequences	IBPSO algorithm
[35]	✓				✓	✓	✓		Particle swarm optimizer
[36]	✓	✓	✓			✓		WGAN-GP	MISOCP by CPLEX solver
[21]	✓				✓	✓		Interval robust set	MICP and by CCG and solver
[25]	✓					✓		FDA	Robust-based IGDT
[31]	✓		✓			✓		Monte Carlo	Markov decision learning
[37]	✓		✓			✓	✓	Kantorovich distance	Genetic algorithm
[38]	✓	✓			✓	✓			MILP by CPLEX solver
[39]	✓	✓	✓	✓		✓			Decomposed MILP and MISOCP subproblems
[40]	✓				✓			LHS	Robust optimization
This paper	✓	✓	✓	✓	✓	✓	✓	WGAN-GP	APSO, MISOCP with TOPSIS

Acronyms: CCG: Column and constraint generation, Ec.: Economic index, FDA: Flow direction algorithm, IBPSO: Improved binary particle swarm optimization, IGDT: Information gap decision theory, LHS: Latin Hypercube Sampling, MILP: Mixed integer linear programming, PDF: Probability density function, Se.: Security index.

renewable generation and storage resources for the ADN to determine their construction time, siting location, and capacity sizing. A chance-constrained information gap decision model is proposed to handle multi-scale and multi-type uncertainties in ADN planning [13]. In addition to the co-configuration of RES and ESS, the optimal operation of substations' on-load tap changers (OLTC) and network reconfiguration with radial and closed-loop operation topologies are also adapted to maximize the RES hosting capacity of the network [14,15]. Meanwhile, the relevance of DR in the expansion planning modeling is stressed in Ref. [16], and a multi-stage model is proposed to accurately include DR in the joint RES expansion planning problem of ADN.

Besides enhancing the capacity of RES, some researches integrate ADN with EVs. Adoption of EVs expands the application range of RES, but the uncertainty in RES increases the operational burden of ADN. A coordinated optimal planning model for EVCS and RES is proposed to coordinate several planning objectives, including system reliability and investment cost, power loss, and voltage stability [17]. Reference [18] incorporates various investment options, including distributed generation and network reinforcement, and proposes an integrated planning framework to effectively accommodate EVs in ADN. Considering the incorporation of reliability into ADN emphatically, reference [19] proposes an enhanced alternative mixed-integer linear programming model for multistage reliability-constrained ADNP, which overcomes low computational efficiency or oversimplification of heuristic approaches. Unlike the aforementioned studies that consider only static values for generation and load requirements, reference [9] simultaneously considers time-varying profiles of RES production, load, and EVCS demand to address temporal mismatches between them. Based on soft open point (SOP), the voltage control can be applied to mitigate operation costs and voltage violations of RES integrated ADN [20]. And the dispatching of SOP coordinated with ESS has been demonstrated in the literature [21] to enhance the system's flexibility and the maximum PV hosting capacity. Lastly, a comprehensive literature study considering the expansion planning of ADN is provided in Ref. [22].

The representative uncertainty description of RES generation and demand is extremely important for ADN planning, and its accuracy affects the adaptability of the plan. Some research neglects or oversimplifies the RES uncertainties and load variability [23,24], while many researches focus on robust optimization and stochastic optimization in order to deal with the random variables introduced by these uncertainties [25,26]. Robust optimization adjusts the uncertainty interval of new energy and load without establishing a specific probability distribution. To deal with the uncertainties of PV locations and capacities, the proposed allocation problem is formulated as an adaptive robust optimization model with integer recourse variables [27], but the optimization result is too conservative, especially for the generation and load outputs with periodic characteristics.

Another widely used stochastic optimization usually assumes that random variables obey the given probability distribution, and transforms uncertain problems into deterministic ones by sampling and other methods [28]. Specifically, optimal planning of EVCS and RES is presented in Ref. [29] using k-means clustering technique to consider the uncertainties related to renewable power sources and EV demand. Furthermore, reference [30] uses heuristic moment matching method to reduce the complexity of high-dimensional discrete variables. Meanwhile, the Monte Carlo Sampling method is applied in Ref. [31] to construct the joint probability distribution scenarios for stochastic variables. However, its computational cost is greatly increased to achieve satisfactory approximate accuracy of probability distribution. Many studies use adversarial learning to generate scenes in a model-free way [32]. builds a Wasserstein generative adversarial network (GAN) for wind scenario generation, which accurately captures the uncertainty features [33]. proposes a generative model integrated with federated learning and GAN, and demonstrates the effectiveness of the generative model in terms of spatiotemporal correlation.

At present, a majority of studies present ADN planning models considering different flexible resources and optimization methods. Based on the literature survey, the major research gaps can be summarized as follows: 1) Some studies, such as references [10–14,21,25,34,35], only consider the coordinated planning of PV, ESS, and SOP to increase the integration capacity. The demand-side charging guidance and ADN active managements are not discussed sufficiently, which can enhance the carrying capacity by improving voltage quality and power flow distribution. 2) Although references [7], [9], [36] and [49] comprehensively consider various flexible resource characteristics and their coordinated configurations, their optimization processes focus solely on economic objectives without accounting for ADN security indexes with distinct dimensions. Furthermore, the solvers struggle to handle the large number of discrete variables related to integration sites, making it challenging to provide DSOs with more comprehensive and better planning solutions that accommodate diverse preferences. The traditional heuristic algorithms, such as references [34,35,37], have simple and clear solution ideas, but generally take long convergence times and tend to fall into local optima. 3) In ADN planning studies considering the influence of high PV integration, references [7–9,18–20,34,35,37–39] either neglect uncertainty impacts or remain confined to explicit modeling of complex features, with an over-reliance on the precise probability distributions. Reference [32,33] provide the adversarial learning methods. They are prone to gradient vanishing or exploding issues, ensuring reliable ADN operation is difficult when the actual scenario deviates from the preset conditions.

To highlight the contributions of this paper, the features of the

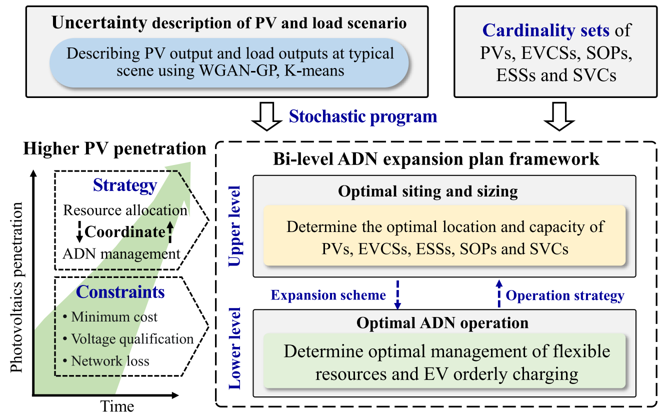

Fig. 1. Framework of the proposed bi-level ADN expansion planning model.

proposed model for ADN planning incorporating flexible resources compared to the previous works are summarized in Table 1. To fill the above research gaps, a bi-level multi-objective coordinated expansion planning method for high-penetration PV integration is proposed. The novelty and main contributions of this paper mainly lies in three parts.

1) This paper cooperatively configures EVCSs with orderly charging guidance, ESSs and active management resources (SOPs, SVCs, and OLTC), aiming to solve the problem of high-penetration PV integration while minimizing cost and keeping flexibility for DNO. The proposed method based on described uncertainties simultaneously determines the siting, sizing, and scheduling of heterogeneous flexible resources in the ADN.
2) A novel bi-level collaborative planning model considering multiple objectives with different attributes such as investment cost, voltage deviation and network loss is proposed. The original non-convex problem is transformed into a solvable mixed-integer second-order cone programming (MISOCP) problem by second-order cone relaxation. And a hybrid algorithm combining adaptive particle swarm optimization (APSO) and SOCP with TOPSIS is proposed to address this problem.
3) An uncertainty description method based on WGAN-GP and K-means for PV generation and load demand is proposed. The Wasserstein loss function and gradient penalty are introduced on traditional GAN model to solve problem of gradient disappearing or explosion, avoiding model assumptions or prior distributions.

with information entropy-based TOPSIS is proposed to solve the mixed-integer second-order cone programming (MISOCP) problem.

The remainder of the paper is organized as follows. Section 2 presents proposed planning model. In Section 3, a detailed method of uncertainty description is introduced. Section 4 presents the case study to evaluate the performance of proposed model. Finally, the conclusions are drawn in Section 5.

2. Modeling and problem formulation

The objective of this study is to facilitate PV integration into an ADN by minimizing the costs for the DNO while maintaining the security and flexibility of the ADN. To address the problem of the lack of PV and load actual scenarios required for ADN planning, an uncertainty description method based on Wasserstein Generative Adversarial Network with Gradient Penalty (WGAN-GP) and k-means is used to generate uncertain scenarios and reduce them to approximate reality. As shown in Fig. 1, the proposed bi-level ADNEP model has two levels. The upper level is focused on optimizing the siting and sizing of flexible resources (e.g., PVs, ESSs, EVCSs, SOPs, OLTCs and SVCs). Based on location and capacity results of upper-level selection, the lower level is focused on optimizing ADN operation and guiding users to charge EVs orderly. The mixed-integer second-order cone programming (MISOCP) problem is solved through adaptive particle swarm optimization with information entropy-based TOPSIS and SOCP method in MATLAB.

2.1. Upper-level problem formulation

The optimization objective of the upper level is to minimize the total investment cost of flexible resources, as introduced in (1). The decision variables include site and capacity selection. The total investment cost comprises the installation costs of the PV, EVCS, ESS, SOP, OLTC, and SVC, which are formulated in Equations (2)–(7).

$$\min C_{ADN} = C_{PV} + C_{EVCS} + C_{ESS} + C_{SOP} + C_{OLTC} + C_{SVC} \tag{1}$$

$$C_{PV} = \sum_{i \in \Omega^{PV}} R \cdot c_{cons}^{PV} \cdot \left(P_i^{cap,PV,MV} + P_{h,i}^{cap,PV,LV} \right) \tag{2}$$

$$C_{EVCS} = \sum_{i \in \Omega^{EVCS}} R \cdot c_{cons}^{EVCS} \cdot P_i^{cap,EVCS} \tag{3}$$

$$C_{ESS} = \sum_{i \in \Omega^{ESS}} R \cdot c_{cons}^{ESS} \cdot P_i^{cap,ESS} \tag{4}$$

$$C_{SOP} = \sum_{i \in \Omega^{SVC}} R \cdot c_{cons}^{SOP} \cdot S_i^{cap,SOP} \tag{5}$$

$$C_{OLTC} = \sum_{i \in \Omega^{OLTC}} R \cdot c_{cons}^{OLTC} \cdot P_i^{cap,OLTC} \tag{6}$$

$$C_{SVC} = \sum_{i \in \Omega^{SVC}} R \cdot c_{cons}^{SVC} \cdot Q_i^{cap,SVC} \tag{7}$$

where R is the capital recovery factor and is derived from the discount rate r, and service life ∂, of Equation (8).

$$R = \frac{r \cdot (1+r)^{\partial}}{(1+r)^{\partial} - 1} \tag{8}$$

2.2. Lower-level problem formulation

The multiple objectives of the lower level are to minimize the annual operational cost and voltage deviation as demonstrated in (9) and (10). Including the voltage deviation of n_Ω nodes as another optimization objective should help ensure the security of the ADN in different PV and load scenarios.

The lower-level multiple objectives include annual operational costs, network loss, and voltage deviation. Due to inconsistent dimensions, they are described separately. The annual operational cost includes power purchased from the superior network, PV curtailment, and maintenance costs, which are calculated by Equation (9). The network losses and voltage deviation are calculated by Equations (10) and (11).

$$\min C_{opt}^{a} = C_b^a + C_p^a + C_m^a \tag{9}$$

$$\min V_d = \frac{\sum_{s \in S} \sum_{t=1}^{T} \sum_{i \in \Omega} \left| V_{s,i,t} - V_{mean} \right|}{S \cdot T \cdot n_\Omega} \tag{10}$$

$$\min l_{loss}^{a} = \sum_{s \in S} \sum_{t=1}^{T} \left[\sum_{l \in \Omega_l^{ij}} \left(r_l \cdot I_{s,l,t}{}^2 \right) \right] \triangle t \tag{11}$$

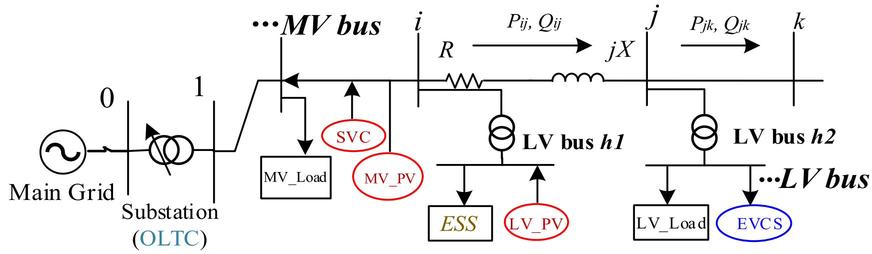

Fig. 2. Integration of flexible resources into an ADN.

$$\begin{cases} C_b^a = \sum\limits_{s\in S}\sum\limits_{t=1}^{T}\left[P_{s,t}^{Sub}\cdot\left(\left(1-u_{sub,s,t}\right)\cdot c_e + c_b\cdot u_{sub,s,t}\right)\right] \\ C_p^a = c_p\cdot\sum\limits_{s\in S}\sum\limits_{t=1}^{T}\sum\limits_{i\in\Omega^{PV}}\begin{bmatrix}\gamma_s\cdot\left(P_i^{cap,PV,MV}\cdot a_{s,t}^{PV} - P_{s,i,t}^{PV,MV}\right) \\ +\left(P_i^{cap,PV,LV}\cdot a_{s,t}^{PV} - P_{s,i,t}^{PV,LV}\right)\end{bmatrix} \\ C_m^a = m_{pv}^a\cdot P^{cap,PV} + \sum\limits_{i\in\Omega^{EVCS}} m_{EVCS}^a\cdot P_i^{cap,EVCS} + \sum\limits_{i\in\Omega^{SOP}} m_{SOP}^a\cdot S_i^{cap,SOP} \\ \sum\limits_{i\in\Omega^{ESS}} m_{ESS}^a\cdot P_i^{cap,ESS} + \sum\limits_{i\in\Omega^{OLTC}} m_{OLTC}^a\cdot P_i^{cap,OLTC} + \sum\limits_{i\in\Omega^{SVC}} m_{SVC}^a\cdot Q_i^{cap,SVC} \end{cases} \quad (12)$$

2.3. Power flow constraints

In this study, the conic form of the power flow equation was adopted because it can realize a more precise model than the DC power flow model and represents the power flow more simply but reasonably than the full AC power flow model. Fig. 2 presents a conceptual illustration of the power flow and the integration sites of flexible resources on both the MV and LV buses. According to Ref. [41], the general formulation of the branch flow model-based optimal power flow (BFM-OPF) is employed in this paper. The following constraints (13)-(14) ensure the active and reactive power balances at each MV bus in different scenarios.

$$P_{s,j,t}^G - P_{s,j,t}^L = \sum_{l\in\Omega_l^{jk}} P_{s,l,t} - \sum_{l\in\Omega_l^{ij}}\left(P_{s,l,t} - {I_{s,l,t}}^2 r_l\right) + g_j {V_{s,j,t}}^2, \\ \forall j\in\Omega, \forall s\in S, \forall t\in T \quad (13)$$

$$Q_{s,j,t}^G - Q_{s,j,t}^L = \sum_{l\in\Omega_l^{jk}} Q_{s,l,t} - \sum_{l\in\Omega_l^{ij}}\left(Q_{s,l,t} - {I_{s,l,t}}^2 x_l\right) + b_j {V_{s,j,t}}^2, \\ \forall j\in\Omega, \forall s\in S, \forall t\in T \quad (14)$$

The available power of each MV bus is decomposed into the injected power of the substation, PV power directly integrated into the MV bus, and excess power injected from LV buses as in Equation (15):

$$P_{s,i,t}^G = P_{s,t}^{Sub} + P_{s,i,t}^{PV,MV} + P_{s,i,t}^{SOP} + \sum_{h\in\Omega_h^i} {P_{s,h,i,t}}^{LV+}, \\ \forall h\in\Omega_h^i, \forall i\in\Omega, \forall s\in S, \forall t\in T \quad (15)$$

$$Q_{s,i,t}^G = Q_{s,t}^{Sub} + Q_{s,i,t}^{SOP} + Q_{s,i,t}^{SVC}, \forall h\in\Omega_h^i, \forall i\in\Omega, \forall s\in S, \forall t\in T \quad (16)$$

The following constraint (17) emphasizes that the system load includes inelastic demand and the power deficiency of LV buses.

$$P_{s,i,t}^L = P_{s,i,t}^{L,MV} + \sum_{h\in\Omega_h^i} {P_{s,h,i,t}}^{LV-}, \forall h\in\Omega_h^i, \forall i\in\Omega, \forall s\in S, \forall t\in T \quad (17)$$

The constraint of node voltage and line current can be expressed by the following Equations (18) and (19).

$${V_{s,j,t}}^2 = {V_{s,i,t}}^2 - 2\left(P_{s,l,t} r_l + Q_{s,l,t} r_l\right) + {I_{s,l,t}}^2\left({r_l}^2 + {x_l}^2\right), \\ \forall l\in\Omega_l^{ij}, \forall i,j\in\Omega, \forall s\in S, \forall t\in T \quad (18)$$

$${I_{s,l,t}}^2 = \frac{{P_{s,l,t}}^2 + {Q_{s,l,t}}^2}{{V_{s,j,t}}^2}, \forall l\in\Omega_l^{ij}, \forall j\in\Omega, \forall s\in S, \forall t\in T \quad (19)$$

The original BFM-OPF is a non-convex nonlinear programming model. Let $\widetilde{V}_{s,i,t} = V_{s,i,t}^2$, $\widetilde{I}_{s,l,t} = I_{s,l,t}^2$. Equations (13), (14) and (18) are transformed to the following Equations:

$$P_{s,j,t}^G - P_{s,j,t}^L = \sum_{l\in\Omega_l^{jk}} P_{s,l,t} - \sum_{l\in\Omega_l^{ij}}\left(P_{s,l,t} - \widetilde{I}_{s,l,t} r_l\right) + g_j \widetilde{V}_{s,j,t}, \\ \forall j\in\Omega, \forall s\in S, \forall t\in T \quad (20)$$

$$Q_{s,j,t}^G - Q_{s,j,t}^L = \sum_{l\in\Omega_l^{jk}} Q_{s,l,t} - \sum_{l\in\Omega_l^{ij}}\left(Q_{s,l,t} - \widetilde{I}_{s,l,t} x_l\right) + b_j \widetilde{V}_{s,j,t}, \\ \forall j\in\Omega, \forall s\in S, \forall t\in T \quad (21)$$

$$\widetilde{V}_{s,j,t} = \widetilde{V}_{s,i,t} - 2\left(P_{s,l,t} r_l + Q_{s,l,t} r_l\right) + \widetilde{I}_{s,l,t}\left({r_l}^2 + {x_l}^2\right), \\ \forall l\in\Omega_l^{ij}, \forall i,j\in\Omega, \forall s\in S, \forall t\in T \quad (22)$$

The effectiveness of the second-order cone relaxation (SOCR) has been thoroughly studied in Ref. [42], which proves that for most ADN structures, SOCR is strictly accurate when the objective function strictly increasing. This paper analyzes this relaxation error in case study to demonstrate the precision of SOCR. Equation (19) is transformed to Equation (23) by SOCR.

$$\left\| \begin{matrix} 2P_{s,l,t} \\ 2Q_{s,l,t} \\ \widetilde{I}_{s,l,t} - \widetilde{V}_{s,j,t} \end{matrix} \right\|_2 \le \widetilde{I}_{s,l,t} + \widetilde{V}_{s,j,t}, \forall l\in\Omega_l^{ij}, \forall j\in\Omega, \forall s\in S, \forall t\in T \quad (23)$$

In addition, the following constraint ensures that the node voltage and line current remain within the allowable upper and lower limits, as shown in Equation (24) and (25).

$$\widetilde{V}_{\min} \le \widetilde{V}_{s,i,t} \le \widetilde{V}_{\max}, \forall i\in\Omega, \forall s\in S, \forall t\in T \quad (24)$$

$$\widetilde{I}_{\min} \le \widetilde{I}_{s,l,t} \le \widetilde{I}_{\max}, \forall l\in\Omega_l^{ij}, \forall s\in S, \forall t\in T \quad (25)$$

The following Equations (26) and (27) ensure a power balance between the LV bus and MV/LV power transmission. The bidirectional transmission power cannot exceed the rated power of the transformer.

$$P_{s,h,i,t}^{PV,LV} + P_{s,h,i,t}^{ESS,dis} - P_{s,h,i,t}^{L,LV_{other}} - P_{s,h,i,t}^{EVCS,LV} - P_{s,h,i,t}^{ESS,ch} = {P_{s,h,i,t}}^{LV+} - {P_{s,h,i,t}}^{LV-}, \\ \forall h\in\Omega_h^i, \forall i\in\Omega, \forall s\in S, \forall t\in T \quad (26)$$

$${P_{s,h,i,t}}^{LV+} \le TR_{h,i}^{lim}\cdot u_{s,h,i,t}^{LV}, {P_{s,h,i,t}}^{LV-} \le TR_{h,i}^{lim}\cdot\left(1 - u_{s,h,i,t}^{LV}\right) \\ \forall h\in\Omega_h^i, \forall i\in\Omega, \forall s\in S, \forall t\in T \quad (27)$$

2.4. Constraints of multiple flexible resources

In this paper, flexible resources are distributed across the network, load, and energy storage sides according to their physical locations. On the network side, the power flow paths can be flexibly adjusted by configuring soft open points (SOPs). The on-load tap changer (OLTC) regulates output voltage, and the static var compensators (SVCs) provide rapid response to voltage changes and adjust reactive power. On the load and energy storage sides, the electric vehicle charging stations (EVCSs) and energy storage systems (ESSs) can smooth out active power fluctuations respectively by guiding orderly charging and balancing time-based differences in power consumption, while simultaneously reducing system operating costs. The aforementioned flexibility resources play crucial roles in different performance of ADN. ESS and EVCS are primarily focused on mitigating fluctuations in power levels, while SVC, OLTC, and SOP are concentrated on reactive power voltage control and power flow regulation.

In the proposed multi-objective planning model, the location, capacity, and operational strategies of each resource are optimized from the DNO's perspective and input into the optimization model. By setting different objective preferences, the model can reflect the role and configuration of flexible resources with different characteristics. Through their coordinated operation, the total stability and flexibility of ADN can be effectively improved to accommodate the high-proportion PV integration.

To simplify the proposed bi-level ADN planning model, the influence of meteorological and geographical differences on the ADN is ignored. The limit of PV generation is expressed in Equations (28) and (29).

$$0 \le P_{s,i,t}^{PV,MV} \le a_{s,t}^{PV} \cdot P_i^{cap,PV,MV}, \forall i \in \Omega^{PV}, \forall s \in S, \forall t \in T \tag{28}$$

$$0 \le P_{s,h,i,t}^{PV,LV} \le a_{s,t}^{PV} \cdot P_{h,i}^{cap,PV,LV}, \forall h \in \Omega_h^{i,PV}, \forall i \in \Omega^{PV}, \forall s \in S, \forall t \in T \tag{29}$$

The integrated PV capacity of the ADN is determined by the total installed PV capacity from the MV and LV buses as in Equations (30)–(32).

$$P_i^{cap,PV,MV} = P_{PV\_rated} \cdot n_{MV,PV,i}, i \in \Omega^{PV} \tag{30}$$

$$P_{h,i}^{cap,PV,LV} = P_{PV\_rated} \cdot n_{LV,PV,i,h}, h \in \Omega_h^{i,PV} \tag{31}$$

$$P^{cap,PV} = \sum_{i \in \Omega^{PV}} \left(P_i^{cap,PV,MV} + \sum_{h \in \Omega_h^{i,PV}} P_{h,i}^{cap,PV,LV} \right), i \in \Omega^{PV}, h \in \Omega_h^{i,PV} \tag{32}$$

PV systems with different capacities were assumed to have the same trend of generation for a given scenario. It is ensured that the reverse power to the upper network is less than the capacity of transformer in Equation (33).

$$P_{h,i}^{cap,PV,LV} \le \rho \cdot TR_{h,i}^{lim}, i \in \Omega^{PV}, h \in \Omega_h^{i,PV} \tag{33}$$

For the ADN planning problem oriented to improve PV penetration, it is necessary to ensure that the requirement issued by the management department of grid needs to be satisfied. The following constraint expresses penetration as the ratio of the total PV integration to the peak load.

$$\zeta \ge \frac{P^{cap,PV}}{\sum_{i \in \Omega} P_i^{p\_load,MV} + \sum_{h \in \Omega_h^i} P_{h,i}^{p\_load,LV}} \times 100\% \tag{34}$$

The SOP can control active and reactive power of ADN to balance the load across different feeders or locations, preventing line overload. A flexible interconnection model based on back-to-back soft open point (SOP) is adopted to flexibly control the active power transmitted between two feeders, and its operation meet the following Equations (35) and (36).

$$\begin{aligned} & P_{s,i,t}^{SOP} + P_{s,j,t}^{SOP} + P_{s,l,t}^{SOP,loss} = 0, \\ & \forall i,j \in \Omega^{SOP}, \forall l \in \Omega_l^{ij}, \forall s \in S, \forall t \in T \end{aligned} \tag{35}$$

$$\begin{aligned} & P_{s,l,t}^{SOP} = A_i^{SOP} \left| P_{s,i,t}^{SOP} \right| + A_j^{SOP} \left| P_{s,j,t}^{SOP} \right|, \\ & \forall i,j \in \Omega^{SOP}, \forall l \in \Omega_l^{ij}, \forall s \in S, \forall t \in T \end{aligned} \tag{36}$$

SOP capacity constraints are expressed in Equations (37)–(39).

$$\sqrt{\left(P_{s,i,t}^{SOP}\right)^2 + \left(Q_{s,i,t}^{SOP}\right)^2} \le S_i^{cap,SOP}, \forall i \in \Omega^{SOP}, \forall s \in S, \forall t \in T \tag{37}$$

$$\sqrt{\left(P_{s,j,t}^{SOP}\right)^2 + \left(Q_{s,j,t}^{SOP}\right)^2} \le S_j^{cap,SOP}, \forall j \in \Omega^{SOP}, \forall s \in S, \forall t \in T \tag{38}$$

$$S_i^{cap,SOP} = n_{sop,i} \cdot S_{SOP\_rated}, \forall i \in \Omega^{SOP} \tag{39}$$

The load of EV charging on each LV bus h can be derived in a similar manner. The same type of EVCS was assumed throughout the ADN. Considering the different charging habits of EV users and the different deployment scenarios in working and residential districts, the EV charging load of each node is determined by the installation number of LV bus h. It can flexibly respond to DNO guidance, and the adjustment interval of charging power is set as ±20 %.

$$P_{h,i,t}^{EVCS} = a_{s,k,t}^{EVCS} \cdot \sum_{k \in \Omega_k^i} n_{EVCS,k,h,i}, \forall i \in \Omega^{EVCS}, \forall s \in S, \forall t \in T \tag{40}$$

$$P_i^{cap,EVCS} = P_{EVCS\_rated} \cdot \sum_{h \in \Omega^{EVCS}} \sum_{k \in \Omega_k^i} n_{EVCS,k,h,i} \tag{41}$$

ESS is primarily used to balance grid load fluctuations, support the PV integration, and provide power balancing services. The power charging and discharging decisions are constrained by Equations (42) and (43), respectively. The ESS state is updated in time steps according to the charging or discharging action, and it has upper and lower limits in Equations (44)–(47).

$$0 \le P_{s,h,i,t}^{ESS,ch} \le n_{ESS,h,i} \cdot \overline{P}_{ESS} \cdot u_{s,h,i,t}^{ESS}, \forall i \in \Omega^{ESS}, \forall s \in S, \forall t \in T \tag{42}$$

$$0 \le P_{s,h,i,t}^{ESS,dis} \le n_{ESS,h,i} \cdot \overline{P}_{ESS} \cdot \left(1 - u_{s,h,i,t}^{ESS}\right), \forall i \in \Omega^{ESS}, \forall s \in S, \forall t \in T \tag{43}$$

$$E_{s,h,i,t}^{ESS} = E_{s,h,i,t-1}^{ESS} + \left(\lambda^{ch} \cdot P_{s,h,i,t}^{ESS,ch} - P_{s,h,i,t}^{ESS,ch} \Big/ \lambda^{dis}\right) \cdot \triangle t, \forall t \ge 1 \tag{44}$$

$$E_{s,h,i,t}^{ESS} = n_{ESS,h,i} \cdot E_{ini}^{ESS}, \text{ if } t = 1 \tag{45}$$

$$n_{ESS,h,i} \cdot (1 - \varepsilon)\overline{E} \le E_{h,i,t}^{ESS} \le n_{ESS,h,i} \cdot \varepsilon \cdot \overline{E}, \forall i \in \Omega^{ESS}, \forall t \in T \tag{46}$$

$$P_i^{cap,ESS} = \overline{E} \cdot \sum_{h \in \Omega_h^i} n_{ESS,h,i}, \forall i \in \Omega^{ESS} \tag{47}$$

OLTC is capable of adjusting the transformer tap voltage in response to load variations without disconnecting the power supply. By changing the tap position of the transformer, it mainly adjusts the voltage on the MV side of an HV/MV bus. The bus node voltage of a substation is converted into an adjustable variable, as shown in Equations (48)–(50).

$$V_{min}^2 \le V_{i,t}^2 \cdot \partial_{s,i,t} \le V_{max}^2, \forall t \in T, \forall i \in \Omega^{OLTC} \tag{48}$$

$$\partial_i^{min} \le \partial_{s,i,t} \le \partial_i^{max}, \forall t, \forall i \in \Omega^{OLTC} \tag{49}$$

$$\partial_{s,i,t} = \partial_i^{min} + \sum_i \partial_i^{min} \cdot u_{s,i,t}^{OLTC}, \forall t \in T, \forall i \in \Omega^{OLTC} \tag{50}$$

The exchanged active power and reactive power through a substation node are limited by the transactive capacity as given in Equations (51) and (52).

$$-S_p^{max} \le P_{s,t}^{Sub} \le S_p^{max}, \forall s \in S, \forall t \in T \tag{51}$$

$$-S_q^{max} \le Q_{s,t}^{Sub} \le S_q^{max}, \forall s \in S, \forall t \in T \tag{52}$$

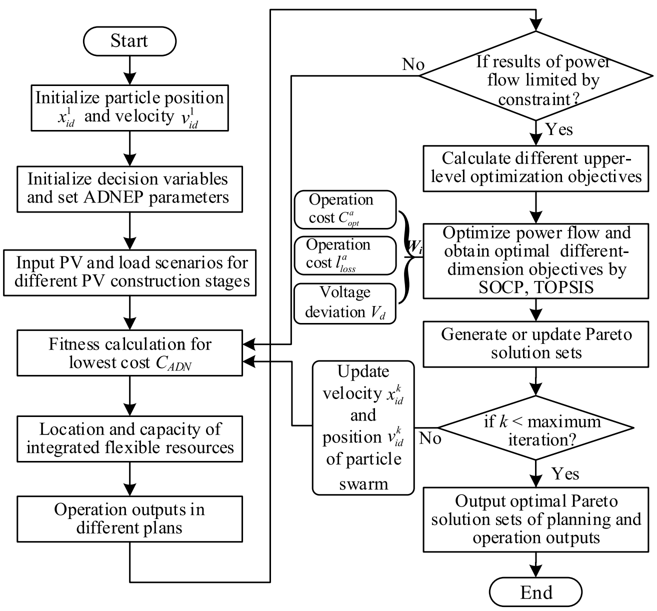

Fig. 3. Solving diagram based on APSO and SOCP for bi-level problem.

The quadratic form of the branch power flow constraints is given by Equation (53).

$$P_{s,l,t}{}^2 + Q_{s,l,t}{}^2 \le S_l{}^2, \forall l \in \Omega_l^{ij}, \forall t \in T \tag{53}$$

SVC possesses fast dynamic regulation capability and can adjust the grid voltage by absorbing or injecting reactive power, thereby enhancing voltage stability. The amount of reactive power compensation is constrained by the upper and lower limits of power compensation.

$$0 \le Q_{i,t}^{SVC} \le Q_i^{cap,SVC}, \forall t \in T, \forall i \in \Omega^{SVC} \tag{54}$$

$$Q_i^{cap,SVC} = n_{SVC,i} \cdot Q_{SVC\_rated}, \forall i \in \Omega^{SVC} \tag{55}$$

2.5. Solution method for bi-level planning problem

The ADN planning problem has bi-level optimization objectives, requiring parameter transmission between upper and lower levels. Due to the multiplication of binary siting and integer sizing variables in this problem, the commercial solvers struggle to find solutions. Therefore, we consider using an Adaptive Particle Swarm Optimization (APSO) algorithm for solving it. Moreover, due to the diverse dimensions of multiple optimization objectives, weighted conversion is essential to obtain appropriate weights. To solve the lower multi-objective scheduling problem, this paper employs an SOCP (GUROBI solver) combined with technique for order preference by similarity to ideal solution (TOPSIS) method, which adjusts the weights of multiple objectives. The detailed solving process is shown in Fig. 3.

Firstly, the inertia weight of APSO is improved adaptively by analyzing the distance between particles and optimal particle. The difference in their position vectors x_{id}^k, was used to guide the value of the inertia weight. In Equation (56), w_i^k value is nonlinearly adjusted according to the gap.

$$w_i^k = w_s - (w_s - w_e)\left(x_i^k - 1\right)^2 \tag{56}$$

where w_i^k is the weight of particle i in iteration k. w_s, w_e are beginning and ending values of the inertia weight respectively.

A dynamic mirror Pareto set is used to update the particle speed and position. In initial distribution stage, a mirror image is established with the optimal particle as an optimal Pareto set. As iteration proceeds, the average value of particles is calculated between each two mirrors, and the optimal solution is the global optimum at this time, so that the speed and position of particles are updated, as shown in Equations (57) and (58).

$$\begin{aligned} v_{id}^{k+1}(t+1) = & w_i^k v_{id}^k(t) + c_1 r_1 \left[p_{id}(t) - x_{id}^{t+1}(t)\right] \\ & + c_2 r_2 \left[\Gamma\left(p_{gd}^k(t)\right) - x_{id}^t(t)\right] \end{aligned} \tag{57}$$

$$x_{id}^{k+1} = x_{id}^k + x_{id}^{k+1} \tag{58}$$

where k is the iteration numbers, v_{id}^k, x_{id}^k is the velocity and position of particle i after iteration k. w_i^k is the inertia weight. c_1, c_2 are the acceleration factor. $r_1, r_2 \in [0, 1]$ as random numbers. p_{gd}^k is the optimal particle of current Pareto sets. $\Gamma\left(p_{gd}^k\right)$ is the superior particle of Pareto sets in two mirrors.

In multi-objective optimization problems, objectives are interdependent and mutually influenced through the variables, and optimizing one objective often comes at the expense of others. Due to the potential inconsistency in evaluation scales across different objectives, it is challenging to directly assess the quality of solutions for this problem. Therefore, the TOPSIS method based on information entropy is used to solve the problem of multi-objective weight allocation in the optimization process. After the normalization, information entropy and weights of objectives to measure uncertainty and diversity of the objectives are calculated by Equations (59) and (60).

$$H(Y) = -\sum_{i=1}^{n} \left(P(y_i)\log_2 P(y_i)\right) \tag{59}$$

$$W_i = \frac{1 - H(y_i)}{\sum_{j=1}^{m}(1 - H(y_i))} \tag{60}$$

where $H(Y)$ represents information entropy of objectives, y_i represents each possible value of objectives, and $P(y_i)$ is the probability of value being y_i. W_i denotes entropy weight of the optimization objective, and m is the total number of objectives.

3. Uncertainty description of PV and load scenarios

An uncertainty description method based on Wasserstein Generative Adversarial Network with Gradient Penalty (WGAN-GP) and K-means was proposed to describe the uncertainties associated with PV generation and load demand. The installed EVCS and PV modules were assumed to remain unchanged, and the load was assumed to increase proportionally in different construction stages. Local historical measurements of meteorological information and the historical load demand were taken as conditional values. When the unsupervised confrontation is completed, data x' are generated. The loss functions of the generator and discriminator [31] are represented by L_G and L_D in Equations (61) and (62), respectively.

$$L_G = E_{x' \sim p\_x'}\{\log[1 - D(x')]\} \tag{61}$$

$$L_D = E_{x \sim p\_x}[\log D(x)] + E_{x' \sim p\_x'}\{\log[1 - D(x')]\} \tag{62}$$

where E is the expectation, $D(\cdot)$ is the output value of the discriminator, and "$\sim$" indicates that the data follow the corresponding probability distribution. Particularly, x and x' separately follow probability distribution p_x and p_x'.

Unlike the GAN model, WGAN-GP can provide a smoother gradient by using the Wasserstein distance instead of the original Jensen-Shannon divergence as the discriminator loss function to measure the distance between the distribution of generated samples and the distri-

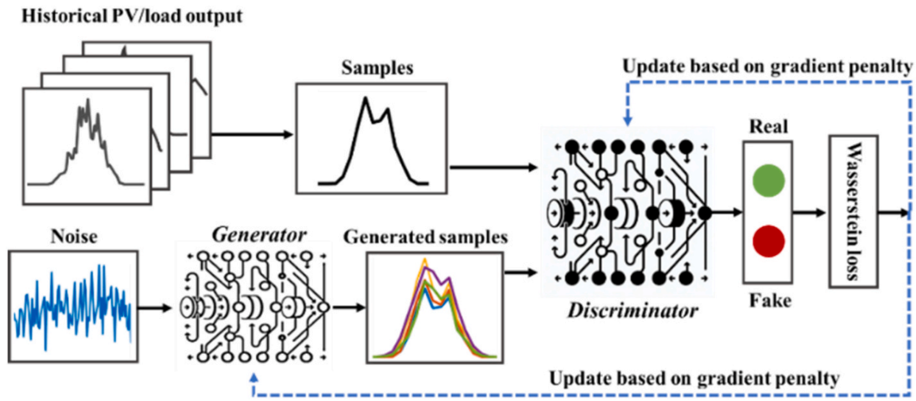

Fig. 4. Basic structure of WGAN-GP.

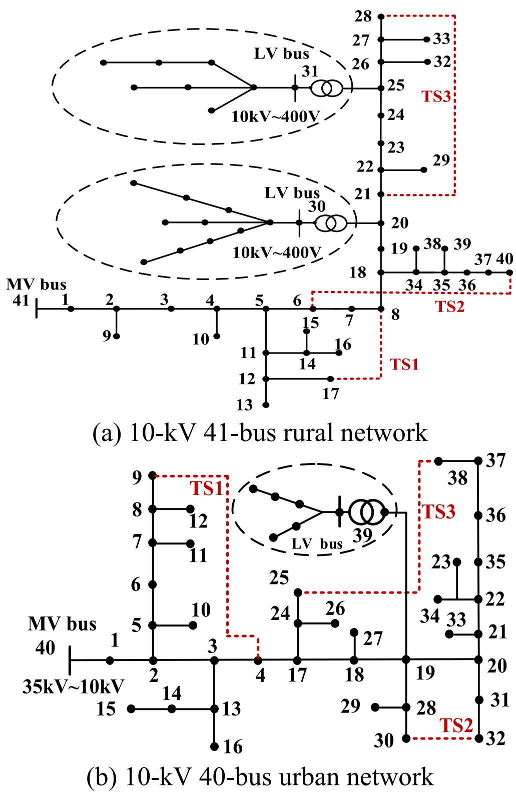

Fig. 5. Distribution network of different characteristics.

bution of real samples. As shown in Fig. 4, the Wasserstein loss can help to improve the stability of training and provide a more useful gradient. Meanwhile, the gradient penalty term ensures that the discriminator

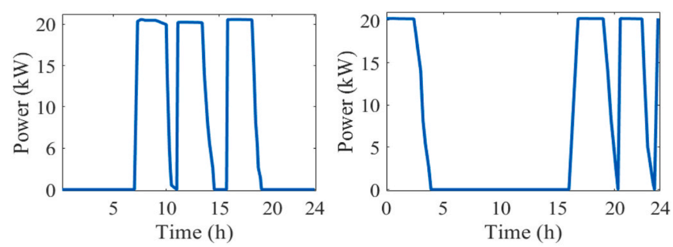

Fig. 6. Typical power demand curves of charging EVs in working or residential areas.

satisfies the 1-Lipschitz constraint, further enhancing training stability. And it does not theoretically require balancing the training level of the generator and the discriminator, thus avoiding gradient vanishing and mode collapse. Therefore, its optimization goal becomes:

$$\min_G \max_D V(G,D) = E_{x\sim p\_x}[\log D(x)] + E_{x^{\cdot}\sim p\_x^{\cdot}}\{\log[1-D(x^{\cdot})]\} \\ -\vartheta E_{\tilde{x}\sim p\_\tilde{x}}\left[(\|\nabla_{\tilde{x}} D(\tilde{x})\|-1)^2\right] \tag{63}$$

where $\|\cdot\|$ represents the 1-norm, $\tilde{x} = \varepsilon x + (1-\varepsilon)G(x^{\cdot})$, and ε obeys a uniform distribution on [0,1], ϑ denotes the regular term coefficient.

To avoid the problem of low efficiency caused by too many scenarios, a k-means clustering algorithm based on Euclidean distance is used to obtain typical scenarios.

4. Case study

4.1. Two test systems, data, and assumptions

To evaluate its effectiveness, the proposed bi-level planning model was applied to two different ADNs in Gansu Province, China: a 10-kV 41-bus rural network with a low load and 10-kV 40-bus urban network with a high load. Differences in the load characteristics and configurable

Table 2
Detailed impedance values of each branch in the 41-bus rural network.

From	To	$Z = R + jX$ (Ω)	From	To	$Z = R + jX$ (Ω)	From	To	$Z = R + jX$ (Ω)	From	To	$Z = R + jX$ (Ω)
41	1	0.67+j0.42	5	11	0.78+j0.49	19	21	0.57+j0.36	25	31	0.16+j0.10
1	2	0.67+j0.42	11	12	0.31+j0.20	20	22	0.05+j0.03	26	32	0.07+j0.05
2	3	0.78+j0.49	12	13	0.21+j0.13	21	23	0.21+j0.13	27	33	2.06+j1.31
3	4	0.21+j0.13	11	14	0.05+j0.03	22	24	0.62+j0.39	18	34	0.52+j0.32
4	5	0.21+j0.13	11	15	0.57+j0.36	23	25	0.05+j0.03	34	35	0.10+j0.07
5	6	0.05+j0.03	14	16	0.31+j0.20	24	26	0.78+j0.49	35	36	0.62+j0.40
6	7	0.93+j0.59	14	17	0.62+j0.39	25	27	0.10+j0.06	36	37	0.21+j0.13
7	8	0.08+j0.04	12	18	0.10+j0.07	27	28	0.36+j0.23	34	38	0.31+j0.20
2	9	0.05+j0.03	8	19	0.26+j0.16	22	29	0.94+j0.59	35	39	0.31+j0.20
4	10	0.05+j0.03	18	20	0.16+j0.90	20	30	0.59+j0.37	37	40	0.10+j0.06

Table 3
Major parameters of the modules(PV, ESS, EVCS, SVC, AND OLTC).

Module	Description	Value
PV	Investment cost of PV unit capacity c_{cons}^{PV} (\$/kW)	856.9
	Unit curtailment cost c_p (\$/kWh)	0.086
	Minimum rated power of installed PV P_{PV_rated} (kW)	20
	Annual maintenance cost m_{pv}^{a} (\$/kW)	12.9
ESS	Investment cost of ESS unit capacity c_{cons}^{ESS} (\$/kWh)	513.6
	Unit energy capacity of ESS modules $\overline{E}$ (kWh)	20
	Charging/discharging efficiency $\lambda^{ch}/\lambda^{dis}$	95 %
	Depth of discharge ζ	90 %
	Annual maintenance cost m_{ESS}^{a} (\$/kWh)	10.3
EVCS	Investment cost of EVCS modules c_{cons}^{EVCS} (\$/kW)	92.9
	Unit capacity of EVCS modules P_{EVCS_rated} (kW)	20
	Annual maintenance cost m_{EVCS}^{a} (\$/kW)	1.9
SOP	Investment cost of unit capacity c_{cons}^{SOP} (\$/kVA)	142.8
	Unit capacity of SOP S_{SOP_rated} (kVA)	100
	Loss coefficient A_i^{SOP}	0.02
	Annual maintenance cost m_{SOP}^{a} (\$/kVA)	2.8
SVC	Investment cost of unit capacity c_{cons}^{SVC} (\$/kVA)	142.8
	Unit capacity of SVC modules Q_{svc_rated} (kVA)	10
	Annual maintenance cost m_{SVC}^{a} (\$/kVA)	2.8
OLTC	Investment cost of ESS unit capacity c_{cons}^{OLTC} (\$/kVA)	62.5
	Annual maintenance cost m_{OLTC}^{a} (\$/kVA)	0.167
Service life of each component ∂		15
Discount rate r		0.065

Table 4
Baseline values of network operational parameters.

Parameter	Description	Value
$V_{\max}/V_{\min}$	Voltage upper and lower limits	0.95/1.05
S_p^{max}/S_q^{max}	Substation capacity limit (MW, Mvar)	3.2/2.1
S_l	Branch flow limit (MVA)	2.9
Electricity prices	1:00–2:00, 5–6:00, 0.5. 3:00–4:00, 10:00–17:00,0.26 7:00–9:00, 18:00–24:00, 0.76	

Table 5
The candidate nodes of flexible resources in the different network.

Candidate sites	41-bus rural network	40-bus urban network
PV	[1–40]	[6,8,10,12,15,18,20,26–33,35,38]
ESS	[6,9–12,15,20–26,28–31]	[4, 10, 14, 18, 22, 32-35, 37, 39]
EVCS	[12, 13, 15–17, 25–40]	[3, 6, 9, 14, 18, 20, 28-33, 36, 39]
SVC	[8–12,25–30]	[9,11,14,16,20,26–28,34,36,37]

resource space of an ADN affect the plan for increasing PV penetration. The urban network had a greater transformer capacity than the rural network. Fig. 5 shows the architectures of the two ADNs. For the rural network, some nodes on the 10-kV main feeder are communal and correspond to 400-V LV buses. The red dashed line represents the set interconnection ties of ADN. The impedance of a branch is calculated based on the pole distance between two nodes according to $Z = 0.650 + j0.412\ \Omega$/km. The impedances of each branch in the 41-bus rural network are exhibited in Table 2. Meteorological information used to describe the trend of PV generation includes the local measured irradiance, temperature, and humidity. The EVCS charging information is provided by the manufacturer. A 20 kW EVCS was selected for both working and residential areas. Fig. 6 shows the typical power demand curves obtained when charging EVs.

Consistent module parameters are adopted, including the investment and maintenance costs of the PV, ESS, EVCS, SVC, and OLTC as well as the capacity of the minimum installed unit, as given in Table 3 [43]. Table 4 presents the baseline values of the operational parameters of the 10-kV networks. The voltage upper and lower limits are set to 0.95 and 1.05 respectively. The candidate nodes of flexible resources are shown in Table 5.

The WGAN-GP algorithm is run in Keras environment of TensorFlow 2.1 and all other algorithms are run on MATLAB R2018. The algorithms are installed on a computer with an Intel Core i7-8700 CPU operating at 3.20 GHz.

4.2. Results analysis of uncertainty description

For the uncertainty analysis, the model is trained using 1 months of limited PV output information from a PV plant in western China as a testing dataset with samples comprising 24-h PV outputs in 1-h intervals. Fig. 7(a) shows the generated samples. The thin lines are power output curves generated by the WGAN-GP to reflect the PV time-series characteristics. PV output is usually characterized according to the weather [44]. The k-means clustering algorithm is used to obtain typical scenarios for different types of weather, as shown by the thick lines in Fig. 7(a). While the PV output depends on the weather, building loads (including EVCSs) often exhibit periodic characteristics that depend on the working state. During nonworking hours, the load trend is more complex and variable, which affects the GAN result. The k-means clustering algorithm is used to reduce the number of generated samples and improve the correlation to real load demand scenarios.

The proposed uncertainty description method is verified by comparison with Latin hypercube sampling (LHS), which generates new scenarios by grouping random sampling. Fig. 8 shows the Taylor diagram comparing the performances of the two methods in terms of the heuristic distance with the corresponding observation [45]. The method based on WGAN-GP outperforms traditional GAN in terms of SD and correlation when describing PV uncertainty, by 0.03 and 2.9 % respectively.

4.3. Planning results description

The planning process is different for the two ADNs owing to the large difference in resource endowment and transformer capacity. In this part, we select a stage $P^{cap,pv} \geq 3.0$ MW to describe the coordinating plan of siting and sizing. To meet the integrated PV capacity requirement and consider the actual charging demand of EV users, ten PV and three EVCS candidate nodes are selected for the rural network. Meanwhile, five PV and four EVCS candidate nodes are selected for the urban network.

Fig. 9 shows the optimization results of the two networks. For the rural network (high-PV and low-load situation), some transformers had a small capacity; hence, a single node had a small PV integration capacity. The largest PV integration capacity is 600 kW. The total integrated PV capacity is 3.1 MW, and the PV penetration rate reaches 124 %, as shown in Fig. 9(a). The EVCSs are deployed at nodes 32, 36, and 39 with a total installed capacity of 240 kW. OLTC is installed at node 41 with a capacity of 6 MVA. The SOP location and capacity are shown in Table 6. For the rural network, charging and discharging by the ESSs are mainly used to stabilize fluctuations in PV generation. Rural networks relying solely on ESSs and orderly charging guidance find it challenging to support high-penetration PVs. SOPs flexibly balance the power flow between feeders to reduce the impact of PV and fluctuations on feeders and node voltage collaborated with OLTC. Fig. 10(a) shows the optimization results of the rural network in sunny weather. From 10:00 to 17:00, PV generated enough power to not only meet the load demand of the rural network but also send a large amount of power back to the upper network. The ESSs installed at different sites have different charging behavior, but through the superimposed bar chart of all ESSs, they generally charge during the daytime and discharge at night.

In contrast to the rural network (high-PV and high-load situation), the 40-bus urban network has a larger transformer capacity. Deploying EVCSs in the urban network has a positive effect on PV integration. The orderly charging behavior of EVCSs is considered from the perspective of the DNO. Fig. 9(b) shows the optimal siting and sizing results of the urban network. The installed PV capacity was 3.25 MW, and PV

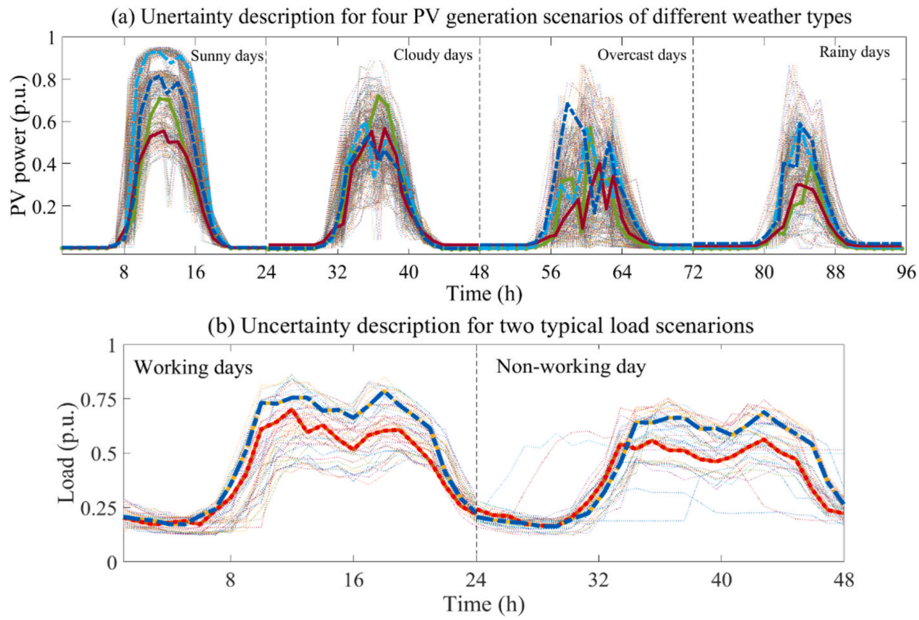

Fig. 7. Uncertainty of PV generation and total load demand.

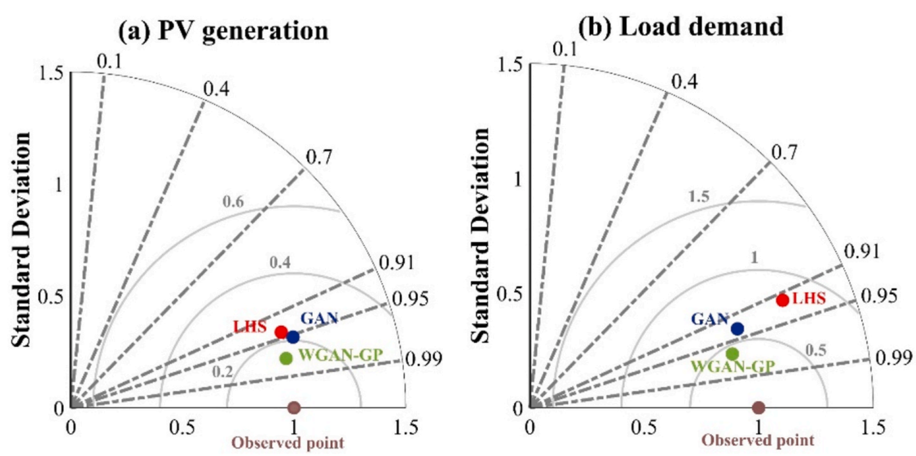

Fig. 8. Taylor diagrams comparing correlations of proposed uncertainty description method (WGAN-GP, green dot) and benchmark method (GAN, blue dot; LHS, red dot) against the observed point (brown dot).

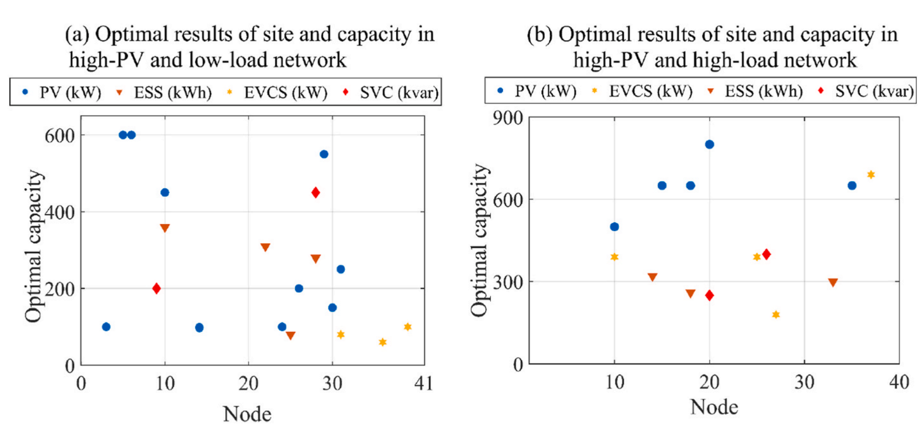

Fig. 9. Optimal siting and sizing results of different networks.

Table 6
Locations and capacities of SOP.

Case	41-bus ADN			40-bus ADN		
SOP location	TS1	TS2	TS3	TS1	TS2	TS3
Capacity (kVA)	300	300	/	200	/	300

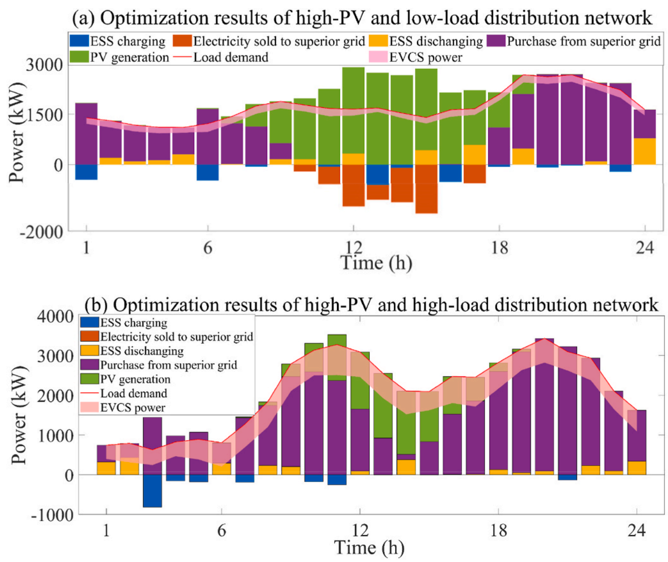

Fig. 10. Daily optimal operation results in different distribution network: (a) rural distribution network, (b) urban distribution network.

Table 7
Performance index of ADN investment and annual operational cost with different planning (A) 41-bus rural network ($\zeta = 1.2$).

Planning Solution	Total cost(M$)	Investment cost(M$)	Operational cost(M$)	PV curtailment (k$)
Case A	14.93	0.91	14.02	50.43
Case B	13.19	0.84	12.35	53.85
Case C	14.15	0.83	13.32	45.30
Case D	12.78	0.77	12.01	48.12
(B) 40-bus urban network ($\zeta = 1.2$)				
Planning Solution	Total cost(M$)	Investment cost(M$)	Operational cost(M$)	PV curtailment (k$)
Case A	25.36	0.93	24.43	50.25
Case B	23.01	0.78	22.23	45.86
Case C	23.58	0.85	22.73	36.12
Case D	22.18	0.75	21.43	38.44

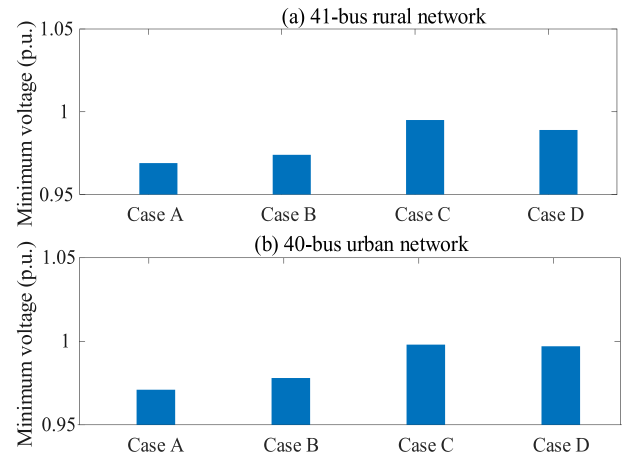

Fig. 12. Voltage deviation and minimum voltage values abstained for both feeders.

penetration reached 104.8 %. The optimal EVCS locations are nodes 10, 26, 28, and 37 with a total capacity of 1.6 MW. Fig. 10(b) shows the optimal operation results of the urban network on an overcast day. Besides improving the economy of ADN operation, the safety objective of the voltage deviation and network loss is to ensure that ESSs complete their charging at night. In this paper, the DNO's response not only meets the rigid demand for EV charging, but also regulates 20 % of the set charging load. During the period of daytime from 9:00 to 18:00, the EV charging power increases, which consumes the excess PV power generation during the noon period, thereby alleviating the overvoltage problem. The guidance of orderly charging also plays an important role in the improvement of PV integration.

4.4. *Case study of different planning strategy*

On the premise of meeting the PV penetration of ADN, the following cases are considered to compare the performance of the proposed model with other typical planning strategies in terms of investment cost, operational cost, and ADN security.

Case A. Consider the ADN planning including the construction of PV, EVCS and ESS;

Case B. An LHS-based uncertainty description method and consider the construction of PVs, ESSs, EVCSs and active management resources (SOPs, SVCs, OLTC);

Case C. A robust optimization method that fully considers the PV output and load scenarios in extreme weather, consider the construction of PVs, ESSs, EVCSs and active management resources (±20 interval);

Case D. (Proposed method): A WGAN-GP-based uncertainty description method, and consider construction of PVs, ESSs, EVCS, SVCs, OLTC

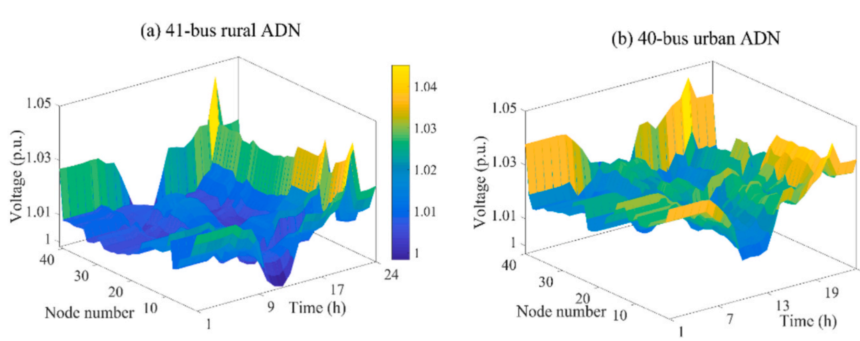

Fig. 11. Voltage distribution in different ADN under overcast condition.

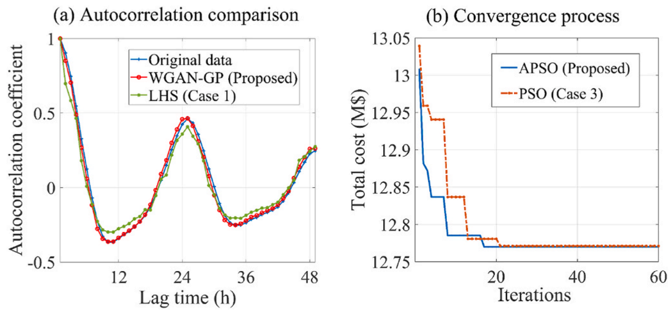

Fig. 13. Comparison of autocorrelation and convergence process.

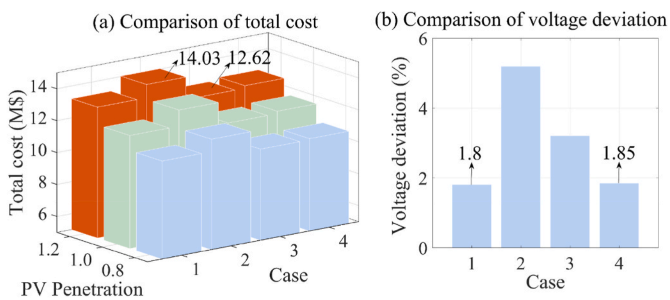

Fig. 14. Comparison of total cost and voltage deviation in different cases.

and SOPs.

The PV integration capacity is set to exceed 3 MW. Table 7 presents the results of the different planning strategies for the two networks. With Case A, In Case 1, a larger-capacity ESS needs to be configured to compensate for the lack of voltage regulation capability, and due to the high unit cost of ESS, the investment costs actually increase instead of decreasing. And the PV curtailment is increased by 4.8 % compared with Case D for the rural network, as shown in Table 7. From the investment cost index, Case C install more flexible resources than the proposed method owing to its consideration of extreme weather conditions. In rural network, although the PV curtailment cost is reduced by 2.82k$ compared with the proposed method, the investment and operational cost are increased by 7.79 % and 10.91 %, and the overall cost is not optimal. With Case D, the investment costs of the rural and urban networks are similar, but the rural network has a lower annual operational cost by 9.42 M$. The proposed method reduces the investment cost and annual operational cost of the rural network by 9.1 % and 2.8 % compared with Case B that using LHS-based method, which demonstrates the advantages of the proposed method.

The voltage deviation and minimum voltage of power flow in different feeders are shown in Fig. 12. Note that for all considered cases, the minimum voltage value of each bus at each time step will not be lower than the predefined minimum voltage, which is 0.95 p.u. In comparison of different plans, the robust plan has more stable minimum voltage, followed by the ADNEP method proposed in this paper. The voltage distribution of rural and urban ADN is shown in Fig. 11, node voltages fluctuate from 0.98 to 1.05 when planning PV and other flexible resources, and the voltage value decreases with the distance from the substation.

To offer a more comprehensive comparison with other similar planning methods, the following experiments are carried out: The improved robust optimization method by the generation of source-load scenarios using LHS to reduce the conservativeness [40] (Case 1); Based on WGAN-CG method, a solving method using SOCR (Gurobi), without considering OLTC and SOPs [36] (Case 2); A multi-objective PSO-based method using iterative power flow calculation (Matpower) [35] (Case 3); The proposed method based on WGAN-CG and the iterative solving process of APSO and SOCP (Gurobi) (Case 4).

Table 8
Comparative analysis of computing efficiency of different cases.

Case	Case 1	Case 2	Case 3	Case 4
Calculation time (h)	1.51	1.67	1.23	1.63

Table 9
ADN planning with different stage of PV penetration in 41-bus rural network.

Parametric index		Planning stages with minimum penetration ζ		
		$\zeta = 0.8$	$\zeta = 1.0$ (increment)	$\zeta = 1.2$ (increment)
Increment of resource capacity	PV (MW)	2.15	0.5	0.45
	EVCS (MW)	0.16	0.03	0.05
	ESS (MWh)	0.80	0.10	0.16
	SVC (MVA)	0.40	0.10	0.15
	OLTC (MVA)	6	/	/
	SOP (MVA)	0.6	0.1	0.2
Increment of investment (M$)		0.57	0.09	0.11
Annual operational cost (M$)		10.21	11.18	12.01
Voltage deviation		1.76 %	1.83 %	1.85 %

Based on the representative scenarios obtained from scenario generation and clustering, autocorrelation coefficients are employed to further compare the WGAN-GP and LHS method. As shown in Fig. 13(a), the autocorrelation curve from WGAN-GP closely matches that of the original series, indicating it captures temporal dependencies more accurately than LHS in Case 1. Under identical experimental settings, the algorithm convergence is also evaluated in Fig. 13(b), which demonstrates that the proposed APSO converges more rapidly than PSO.

The total cost and voltage deviation results are presented in Fig. 14. In Case 1, after reducing conservativeness of robust optimization using LHS, the total cost decreased by 6.7 % compared to the traditional robust method of Case C in Table 7. However, this improvement led to an increase in voltage fluctuation of 1.8 % increase. Compared to the random optimization method proposed in this paper at $\zeta = 1.2$, the cost increases by 0.42 M$, while the voltage deviation is reduced by 0.05 %. In Case 2, the absence of OLTC voltage regulation and SOP overload mitigation results in the regulation cost being absorbed by other resources, such as ESS, leading to worse performance. In Case 3, the fluctuation of PV outputs is not considered, and a higher smoothing factor for PV output is selected. It increases the total power generation and reduces fluctuations under the minimum requirement of PV integration. As a result, the reserve capacity is not needed to address short-term fluctuations, reducing the required ESS capacity. The needs for SVC and SOP capacity also decrease due to smaller voltage and flow fluctuations, resulting in a lower total cost of 0.16 M$ than the proposed method. However, when faced with other PV output scenarios, the average voltage fluctuation in case 3 increases by 1.36 %, as shown in Fig. 14(b).

The computation times are shown in Table 8. In Case 3, the nested power flow calculation based on Matpower, rather than optimization, results in shorter computation time. Compared to Case 1, the time is reduced by 0.12 h due to the absence of neural network training. In Case 2, the solver struggles with the strong nonlinearity caused by the product of discrete site variables and other integer variables, so an outer site combination search is used in this case, increasing the computation time by 0.04 h compared to the proposed method. Despite differences in solution approaches, all cases are solved within a reasonable time for planning tasks that do not require strict real-time requirements.

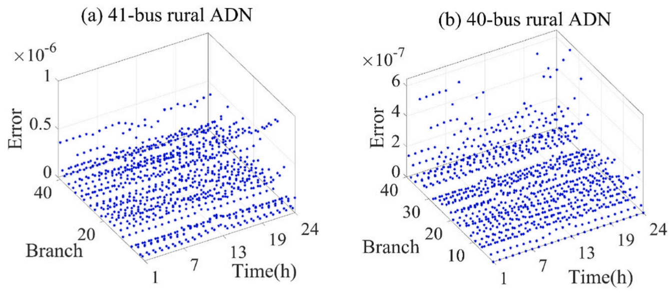

Fig. 15. SOCR error scatter plot.

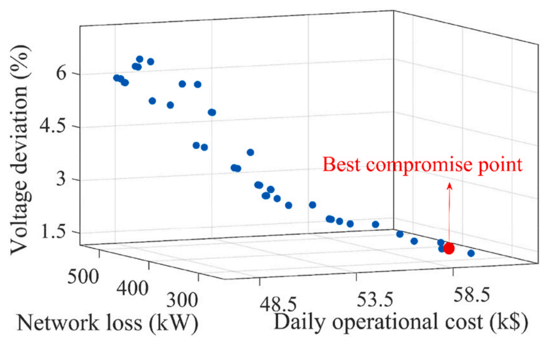

Fig. 16. Pareto optimal solution sets.

4.5. *Case study in different PV penetration*

To evaluate the evolution of resource configuration at different stages of PV penetration, three minimum PV penetration indices were defined: 0.8, 1.0, and 1.2. Table 9 presents the results of the proposed bi-level ADN planning model in different stages of the 41-bus rural network. At $\zeta = 0.8$, the proposed model configured the various flexible resources to ensure a small voltage deviation and small network loss while minimizing the operational costs and providing space for future increments in PV penetration. At $\zeta = 1.0$ and $\zeta = 1.2$, more flexible resources need to be configured, which increased the investment costs and annual operational costs by 0.02 M$ and 0.83 M$, respectively. It can be seen from the voltage deviation that with the gradual increase of PV integration, the voltage fluctuation becomes larger and increases the risk of voltage crossing the limit.

4.6. *Description of SOCR accuracy and multi-objective optimization*

To address the non-convex problems associated with power flow constraints, the proposed model performs a SOCR approximation in terms of the AC power flow constraints. The following calculation formula of SOCR error $\Delta_{ij,t}^{diff}$ is defined:

$$\Delta_{ij,t}^{diff} = U_{i,t} I_{i,t} - \left(P_{ij,t}^2 + Q_{ij,t}^2\right) \tag{64}$$

The relaxation errors of each branch across all time are calculated and presented in Fig. 15. As shown in Fig. 15(a), the maximum relaxation error is on the order of 10^{-6}. Notably, the 41-node network exhibits higher errors than the 40-node distribution network, yet both satisfy the precision requirements and the feasibility of the relaxation approach implemented by SOCR. It also implies that the condition that the SOCR solution is a global optimum of the original problem is basically satisfied. And the results align with reference [36].

Under the same basic parameter settings, this paper analyzes the performance of operational schemes under different optimization objectives, including operational cost, distribution network loss, and voltage deviation. The TOPSIS method is used to find the optimal compromise point between these multi-objectives with different dimensions. The Pareto front is shown in Fig. 16. The Pareto front curve maintains continuity, and the solution set exhibits relatively uniform distribution. The curve clearly demonstrates the relationships among the three objectives. Overall, the operational cost is inversely proportional to the other two security objectives. As ADN network loss or voltage deviation increases, the daily operational cost decreases accordingly, which aligns with practical operation scenarios.

Table 10
Analysis of 41-bus ADN planning results of single and multiple objective.

Results	Daily operational cost (k$)	Network loss (kW)	Voltage deviation
1	45.3	480	6.05 %
2	62.0	313	1.36 %
3	59.8	318	1.76 %

Table 10 presents three typical solutions with a focus on different objective functions. The DNO can choose different options based on their preferences. Among them, Solution 1 focuses on economic benefits, and although its cost is lower than the other solutions, the other indicators are noticeably higher. Solution 2 places more emphasis on ADN network loss and voltage deviation. In this case, the network loss is reduced by 162 kW, and the voltage deviation is reduced by 4.95 %. Solution 3 considers the relationship between the three indicators. Compared to 1, although the operational cost increases, the network loss and voltage deviation are significantly reduced, achieving a higher overall satisfaction and balancing all three indicators.

4.7. *Operational results of multiple flexible resources*

The 41-bus rural ADN is employed to illustrate the supporting roles of multiple flexible resources. The typical PV output scenarios under sunny and cloudy conditions are depicted by the red line in Fig. 7(a). The maximum number of daily voltage adjustments is set to 5 times. Fig. 17 presents the outcomes of OLTC voltage regulation and SVC reactive power compensation. In the early morning when there is no PV power generation, the overall voltage is low. The SVC operates at a higher tap setting while adjusting the OLTC ratio to maintain voltage stability. Around midday, when the PV output reaches its peak, both the OLTC and SVC remain at lower settings to stabilize the voltage, particularly under the sunny scenario, as shown in Fig. 17(a). By contrast, under the overcast condition, the OLTC and SVC generally operate at higher tap positions to compensate for the reduced PV generation. Between 12:00 and 16:00, as the load decreases, the OLTC ratio is lowered accordingly to accommodate voltage changes.

To evaluate the ESS response strategy for flexible regulation requirements, Fig. 18 presents the optimal operational schemes and capacity variations of four ESS units in a 41-bus rural ADN. These units exhibit similar operational patterns and capacity trends. During midday peak PV generation periods, the ESS units charge to prevent overvoltage risk. In early morning and nighttime hours, they compensate for demand-supply imbalances, improving system economic efficiency. Notably, despite low load conditions between 3:00 and 4:00, the ESS units utilize low electricity prices to charge, thereby increasing the overall system load. During this period without PV generation, both reactive power compensation and OLTC operate at elevated settings (as demonstrated in Fig. 17) to ensure ADN stability.

To examine the impact of flexible resources on system voltage regulation, Fig. 19 presents the voltage profiles at nodes 6 and 29 under different PV output scenarios. Under the sunny scenario with only the high-proportion PV generation, the voltage at these nodes rises significantly, posing an overvoltage risk. Once flexible resources are incorporated, the voltage fluctuation is notably reduced. By contrast, under the overcast condition, At the end-of-line node 29, midday PV output is relatively low, while the load is comparatively high, leading to a maximum voltage of 1.027 p.u. Between 1:00 and 7:00, the load is

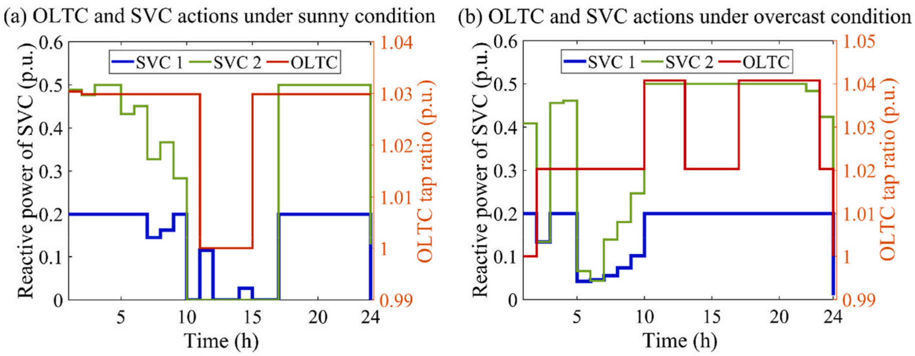

Fig. 17. OLTC and SVC actions under different PV scenarios.

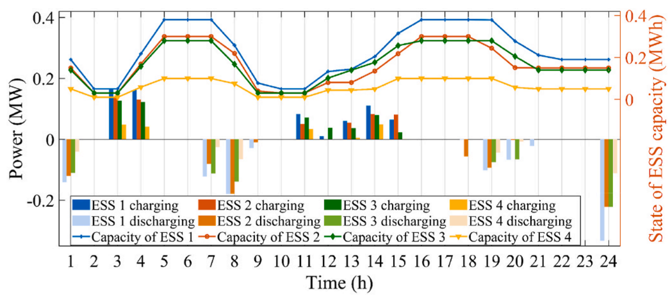

Fig. 18. ESS optimization results in different sites of ADN.

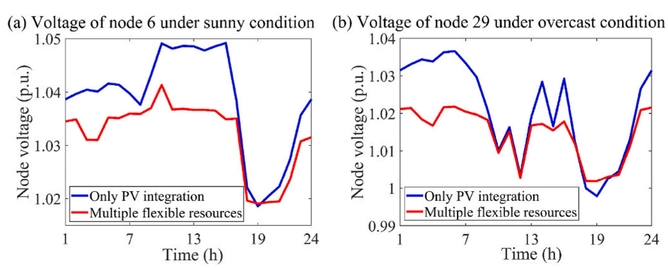

Fig. 19. Node voltage under the integration of multiple flexible resources.

minimal, resulting in lower line currents and reduced voltage drops. Consequently, with a stable upstream voltage, the end-of-line nighttime voltage is higher than at midday. Through the coordinated operation of the OLTC and SVC, voltage stability is further enhanced.

4.8. Validation on the rural 109-bus distribution network

To validate the scalability and effectiveness of the proposed planning method in large-scale distribution networks, this paper also selects an actual rural 109-bus network in Gansu Province, with active and reactive power demands of 16.5 MW and 12.38 Mvar, respectively. Given that this ADN has more branches and a longer power supply radius between line terminals and the substation, which leads to significant voltage drop issues, as shown in Fig. 20. Therefore, a greater number of flexible resources are deployed to mitigate power fluctuations and enhance voltage stability across the complex feeders. Other parameters remain consistent with the 41-node test system.

The optimization results for sites and capacities of flexible resources are shown in Fig. 21. To reduce the search space for location variables, the candidate nodes for PV installation are set on the main feeders. Among these nodes, the one closest to the substation (node 6) hosts the

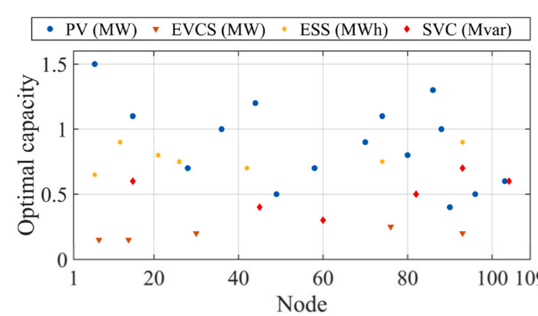

Fig. 21. Optimal siting and sizing results in 109-bus networks.

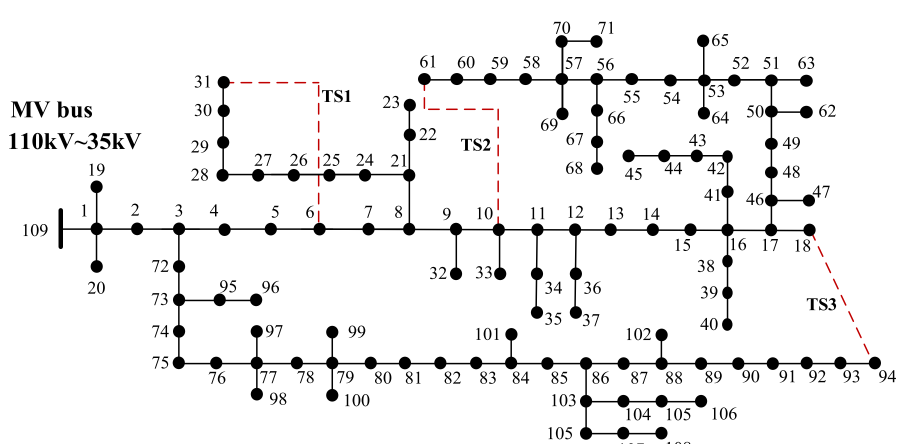

Fig. 20. 109-bus rural network.

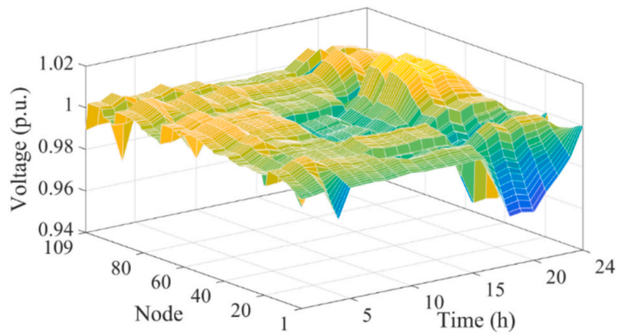

Fig. 22. Node voltages of 109-bus network under sunny condition.

Table 11
Performance index of 109-bus ADN investment and annual operational cost with different plans ($\zeta = 0.8$).

Planning Solution	Total cost(M$)	Investment cost(M$)	Operational cost(M$)	PV curtailment (M$)
Case A	76.49	1.26	75.23	0.20
Case B	75.76	1.67	74.09	0.21
Case C	76.57	1.76	74.81	0.16
Case D	75.28	1.62	73.66	0.18

highest PV capacity at 1.5 MW. Overall, the total PV capacity amounts to 13.3 MW, yielding a penetration level ζ of 80.6 % and fulfilling the first-stage PV deployment requirements. To avoid the pronounced voltage drops at the terminal due to the higher cumulative impedance, SVC with a capacity of 3.1Mvar is integrated. The SOP switches are all in the open state. Through reactive power compensation, the node voltage is maintained within an acceptable range, as shown in Fig. 22. The node voltage distribution is analyzed from the perspective of the DSO's preference for voltage stability. At midday, the coordinated action of OLTC voltage reduction and ESS smoothing effectively prevents any significant voltage increase despite high PV generation, keeping the voltage near its designated standard value. At night, some node voltages hover around 0.96 p.u. The overall voltage fluctuations remain at a relatively low level, demonstrating the effectiveness of the coordinated planning strategy.

Table 11 presents the economic benefits of the proposed planning method under different cases, with settings consistent with Section 4.4. In Case A, although ESS and EVCS help smooth supply-demand fluctuations, in the absence of active voltage regulation, when excessive PV generation drives the voltage to its upper limit, a portion of the power must be curtailed, resulting in an annual curtailment penalty increase of 0.02 M$. Overall, total costs are reduced by up to 1.29 M$ compared to other cases. Despite the increased number of nodes and more complex power flows add to the computational burden, the APSO algorithm, by invoking the Gurobi solver with an optimality tolerance of 0.001, can still obtain the optimal solution. These results demonstrate that the proposed method is adaptable to ADNs of various scales and delivers better performance.

5. Conclusion

In this paper, a bi-level coordinated ADN expansion planning method is proposed to accommodate high-penetration PVs integration while maintaining economic efficiency and flexibility of ADN. An uncertainty description method based on WGAN-GP and k-means is presented to generate uncertain scenarios to approximate reality. Compared to traditional GAN, the correlation index for describing PV uncertainty using presented method has been improved by 2.9 %. Based on described uncertainties, this paper cooperatively configures EVCSs with orderly charging guidance, ESSs and active management resources. To solve this MISOCP problem, a hybrid algorithm combining adaptive particle swarm optimization (APSO) and SOCP with information entropy-based TOPSIS is proposed. The proposed method is evaluated using rural and urban networks, compared to plans without active resource management, total costs of the proposed coordinated plan is respectively decreased by 14.4 % and 12.5 %. It can significantly enhance the economic operation of distribution systems, especially in integration of high penetration of PVs. Although SOP and OLTC installation requires extra budget, the operational cost of ADN can be considerably reduced owing to the flexibility offered by this active management resources.

However, this paper does not consider the three-phase unbalanced power flow that exist in LV networks. The allocation of flexible resources is mainly discussed within the MV network. This issue will be addressed in future research, and the multi-stage planning around the uncertainty of line outages during emergencies will be also investigated.

CRediT authorship contribution statement

Jianjing Li: Writing – original draft, Methodology, Conceptualization. **Fan Li:** Validation, Funding acquisition. **Kai Sun:** Writing – review & editing, Methodology, Data curation. **Bo Sun:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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